Digital Signal Processing Assignment # 1

1. Determine whether each of the following signals is periodic. If it is, find the fundamental period.

   (a) $x(n) = \sin\left(\frac{\pi n}{4} + \frac{\pi}{8}\right)$

   (b) $x(n) = \sin\left(\frac{\pi n}{4} + \frac{\pi}{8}\right) + \cos\left(\frac{\pi}{3}n\right)$

   (c) $x(n) = \sum_{m=-\infty}^{\infty} \delta(n - 2m)$ (hint: try to sketch the signal)

   (d) $x(n) = \sum_{m=0}^{\infty} \delta(n - 2m)$

   (e) $x(n) = \sum_{m=-\infty}^{\infty} (-1)^m \delta(n - m)$

2. Determine if the following signals are energy signal or power signal or neither.

   (a) $x(n) = 2\left(\frac{1}{3}\right)^n$, $n > 0$

   (b) $x(n) = u(n)$

3. Consider a signal, $x(n) = [-3, 1, \underset{\uparrow}{2}, -2, 3, -1]$. Manually sketch the following signals

   (a) $x(n)$

   (b) $x(2 - n)$

   (c) $x\left(\frac{2}{3}n + 1\right)$